

A
Mini Project Report
on
InterviewIQ: AI Mock Interview System
Submitted in fulfillment of the requirements for the degree
Third Year Engineering – Computer Science Engineering (Data Science)

by

Arya Patil	22107047
Nisha Patel	22107037
Prachi Pawar	22107018
Renuka Udugade	22107048

Under The guidance of
Ms. Poonam Pangarkar



DEPARTMENT OF CSE (DATA SCIENCE)
A. P. SHAH INSTITUTE OF TECHNOLOGY
G.B. Road, Kasarvadavali, Thane (W), Mumbai - 400615
UNIVERSITY OF MUMBAI

Academic Year: 2024 – 2025

CERTIFICATE

This is to certify that the Mini Project report on **InterviewIQ: AI Mock Interview System** has been submitted by **Arya Patil (22107047), Nisha Patel (22107037), Prachi Pawar (22107018), Renuka Udugade (22107048)** who are a bonafide students of A. P. Shah Institute of Technology, Thane, Mumbai, as a partial fulfilment of the requirement for the degree in **CSE (DATA SCIENCE)**, during the academic year **2024-2025** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

Ms. Poonam Pangarkar

Guide

Ms. Anagha Aher

Head of Department of CSE Data Science

Dr. Uttam D. Kolekar

Principal

External Examiner:

1.

Internal Examiner:

1.

Place: A. P. Shah Institute of Technology, Thane

Date:

ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide, **Ms. Poonam Pangarkar**. Expressing gratitude towards our HOD, **Ms. Anagha Aher** and the Department of CSE Data Science for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our project coordinator **Ms. Sarala Mary** who gave us her valuable suggestions and ideas when we were in need of them. We would also like to thank our peers for their helpful suggestions.

TABLE OF CONTENTS

Abstract

1. Introduction.....	1
1.1. Purpose.....	1
1.2. Problem Statement.....	2
1.3. Objectives.....	2
1.4. Scope.....	3
2. Literature Survey.....	5
3. Proposed System.....	6
3.1. Features and Functionality.....	6
4. Requirement Analysis.....	8
5. Project Design.....	10
5.1. Use Case diagram.....	10
5.2. DFD (Data Flow Diagram)	11
5.3. System Architecture.....	13
5.4. Implementation.....	14
6. Technical Specification.....	20
7. Project Scheduling.....	23
8. Results.....	25
9. Conclusion.....	28
10. Future Scope.....	29

References

ABSTRACT

The InterviewIQ: AI Mock Interview System is a web-based application designed to help job seekers enhance their interview skills through AI-driven mock interviews. This system utilizes Next.js (React framework) for the frontend, Flask for the backend, and integrates AI models for generating interview questions, analyzing responses, and providing feedback. Users input their job role, description, and experience level, after which the AI generates five tailored interview questions. The system allows users to record their responses, converts speech to text, and analyzes the answers using Natural Language Processing (NLP) techniques. Feedback includes clarity, relevance, sentiment, and confidence scoring, helping users improve their performance. The system also features secure authentication using Clerk and deploys on Vercel for seamless accessibility. By leveraging AI and machine learning, the AI Mock Interview System provides personalized and interactive interview preparation, boosting users' confidence and readiness for real-world job interviews.

Chapter 1

Introduction

The Interview preparation is a crucial step for job seekers aiming to secure their desired roles. Many candidates struggle with answering questions effectively, managing nervousness, and understanding how to improve their responses. Traditional interview coaching methods can be expensive and time-consuming, making it difficult for individuals to practice efficiently.

The AI Mock Interview System is designed to provide a technology-driven solution to these challenges. By utilizing Artificial Intelligence (AI) and Machine Learning (ML), the system generates personalized interview questions based on user input, allowing candidates to practice in a structured, interactive environment. Users can record their answers, which are then analyzed using Natural Language Processing (NLP) techniques to evaluate clarity, relevance, confidence, and sentiment.

This system aims to bridge the gap between theory and real-world interview performance by offering instant AI-driven feedback. Unlike traditional mock interviews, which require human evaluators, this system provides an on-demand practice tool that is accessible, scalable, and cost-effective.

1.1. Purpose

InterviewIQ is designed to provide job seekers with an interactive, AI-driven platform to practice and refine their interview skills. By leveraging machine learning, speech analysis, and NLP-based evaluation, the system generates personalized interview questions, analyzes responses, and offers real-time feedback to assess clarity, relevance, and confidence. Through structured interview simulations, users can enhance their response formulation, articulation, and overall communication skills. The system eliminates the need for human evaluators by providing an objective, data-driven approach to answer evaluation, ensuring accessibility and efficiency. Its primary goal is to serve as a cost-effective, intelligent solution that empowers job seekers to build confidence and improve their interview performance through personalized insights and continuous practice.

1.2. Problem Statement

Many job seekers struggle with interview anxiety, lack of preparation, and difficulty in assessing their own responses. Traditional mock interviews often require the presence of professional interviewers or mentors, making them expensive, time-consuming, and inaccessible to many individuals. Additionally, online resources such as pre-recorded interview guides fail to provide interactive and personalized feedback, leaving candidates uncertain about how to improve their answers.

Moreover, even when candidates practice, they often lack objective evaluation metrics to gauge their performance. Without proper insights into speech clarity, relevance, confidence, and tone, job seekers may continue making the same mistakes without realizing it. A technology-driven solution that integrates AI-powered question generation, real-time speech analysis, and NLP-based feedback can bridge this gap by offering instant, scalable, and data-driven insights for continuous improvement.

1.3. Objectives

The AI Mock Interview System is designed to help job seekers enhance their interview preparation through AI-driven personalized question generation, real-time feedback, and speech analysis. A report on such a system should outline the specific objectives it seeks to achieve. Here are some key objectives to consider:

1. To safeguard user data with advanced authentication -

This is about security. InterviewIQ uses a system called Clerk to make sure your personal information and your recorded interviews are safe. Clerk handles things like logins and passwords, so you can be confident that your data is protected from unauthorized access.

2. To generate personalized interview questions -

Imagine you're applying for a software engineer job. InterviewIQ uses Google's Gemini AI to read the job description and your resume. It then creates practice interview questions that are specifically tailored to that job and your skills. So, instead of getting generic questions, you get questions that are very likely to be asked in your real interview.

3. To deliver actionable insights on answers (feedback) –

After you record your answers, Gemini AI analyzes them. It doesn't just check what you said (content), but also how you said it (tone and delivery). It gives you feedback like: "Your answer was good, but you sounded a little unsure," or "You explained the concept clearly, but try to speak a bit slower." These insights are designed to be helpful and give you specific things to work on.

4. To offer user-friendly platform to record and review mock interviews -

This means InterviewIQ lets you easily record your practice answers (like a video or audio). It stores these recordings in a database (PostgreSQL using Neon), and lets you play them back. This lets you see and hear yourself, so you can analyze your performance. Drizzle is a tool that makes it easy for InterviewIQ to interact with that database.

By clearly defining these objectives, the report can effectively evaluate the success of the AI Mock Interview System and identify areas for improvement.

1.4. Scope

AI mock interview systems use artificial intelligence to simulate real job interviews for practice, feedback, and evaluation. Their scope is broad, encompassing capabilities like analyzing responses, assessing communication and technical skills, making them valuable for job seekers, educational institutions, and companies across various interview types and industries.

1. Job seekers –

An AI mock interview system provides a safe and low-pressure environment where they can practice their interview skills and build their confidence before facing real interviews. The system is useful in helping candidates identify their strengths and pinpoint areas where they need to improve the content and delivery of their interview responses. By familiarizing candidates with the interview process through repeated practice, the system can significantly contribute to reducing anxiety and nervousness associated with real job interviews.

2. Educational institutes –

For educational institutions like universities and career counselling centers, AI mock interview systems offer valuable career development support for students as they prepare for internships and job placements.

3. Training and Development team -

In the realm of training and development, companies can leverage AI mock interview systems to enhance the skills of their existing employees, preparing them for internal promotions or transitions to new roles within the organization. The adaptability of these systems allows them to be used to simulate leadership interviews, providing a tool for assessing and developing the leadership potential of employees.

By outlining these aspects, the report will provide a comprehensive analysis of the AI Mock Interview System, covering its functionality, impact, and potential future developments.

Chapter 2

Literature Survey

The rapid advancement of Artificial Intelligence (AI) has revolutionized various domains, including recruitment and interview preparation. Traditional mock interviews often rely on human evaluators, which can introduce biases and inconsistencies. To address these challenges, researchers have explored AI-driven mock interview platforms that offer objective, real-time feedback to candidates.

The paper titled "An AI Mock-interview Platform for Interview Performance Analysis," authored by [1] Yi-Chi Chou, Felicia R. Wongso, Chun-Yen Chao, and Han-Yen Yu, was presented at the 2022 10th International Conference on Information and Education Technology (ICIET) in April 2022. This research introduces an interview platform designed to assist job-seekers. Job seekers can utilize it for mock interviews to enhance their performance. In our system, the user doesn't have to type the answers which is time-consuming. In fact, the user will give verbal answers which will be converted into text.

Authored by Rubi Mandal, Pranav Lohar, Dhiraj Patil, Apurva Patil, and Suvarna Wagh, the research paper "AI-Based Mock Interview Evaluator" [2] was showcased at the 2023 International Conference on Intelligent Systems for Communication, IoT, and Security (ICISCoIS). The primary goal of this research is to develop an AI-driven mock interview platform that provides objective and automated feedback during interviews. Along with this, our system provides correct answers so that the user gets to know where he/she went wrong.

Focusing on AI-driven interview simulations, the paper "AI-Driven Virtual Mock Interview Development" [3] was written by Prabhat Kumar Mishra, Arunkumar Arulappan, In-Ho Ra, Thanga Mariappan L, Gina Rose G, and Young-Seok Lee, and presented at the 2024 Joint 13th International Conference on Soft Computing and Intelligent Systems and the 25th International Symposium on Advanced Intelligent Systems (SCIS/ISIS) on November 9, 2024..The paper focuses on the development of an AI-driven virtual mock interview platform designed to assist job seekers in enhancing their interview skills. This system does not save user's previous mock interviews whereas our system saves and retrieve user's previous mock interviews from PostgreSQL Database.

Chapter 3

Proposed System

To enhance interview preparation and improve candidates' communication skills, we propose an AI Mock Interview System that leverages Natural Language Processing (NLP), Machine Learning (ML), and Speech-to-Text analysis to create a simulated interview environment. By generating industry-specific interview questions, analyzing spoken responses, and providing AI-driven feedback, the system offers a structured, data-driven approach to refining users' articulation, confidence, and response quality. Unlike traditional mock interviews that rely on human evaluators, this automated and scalable solution ensures accessibility across various industries and job roles, making interview practice more efficient, objective, and personalized for job seekers of all experience levels.

3.1. Features and Functionality

A well-designed AI Mock Interview System can significantly enhance user experience, increase sales, and drive customer loyalty. By leveraging advanced algorithms and data analytics, such systems can offer tailored product suggestions based on individual preferences, purchase history, and browsing behavior. Given below are some features you can expect in our recommendation system:

1. AI-Generated Interview Questions -

The system dynamically generates interview questions based on user input, such as job role, job description, and years of experience. This ensures a personalized and industry-relevant interview experience.

2. Speech-to-Text Conversion -

The system records and converts spoken responses into text, allowing for detailed analysis of verbal communication, including clarity, pronunciation, and pacing.

3. AI-Powered Answer Evaluation -

User responses are analyzed using NLP and ML algorithms to assess factors like relevance, coherence, tone, and confidence, providing structured feedback for improvement.

4. Real-Time Feedback System -

The platform provides instant feedback, highlighting strengths and suggesting improvements in content delivery, clarity, and professional articulation.

5. Performance Tracking and Analytics -

Users can review their past interviews, track their improvement over time, and analyze their progress through performance metrics and insights.

6. Secure Authentication and Data Privacy -

The system integrates Clerk authentication to ensure that user data remains private and secure, with access limited to the respective user.

By incorporating these cutting-edge AI technologies, the proposed system offers a structured, efficient, and personalized interview practice solution that enhances users' preparedness and confidence for real job interviews.

Chapter 4

Requirement Analysis

Requirement analysis is a crucial step in the development of the AI Mock Interview System, ensuring that the platform meets user expectations while maintaining efficiency, accuracy, and security. The requirements are categorized into functional and non-functional aspects, each addressing different components of system functionality and performance.

Functional Requirements

Functional requirements define the core operations of the AI Mock Interview System, specifying what the system must do to fulfill its intended purpose

- **User Registration and Authentication** The system must provide secure user registration and authentication using Clerk and Google sign-in to ensure authorized access.
- **Customizable Interview Parameters** – Users should be able to specify job roles, experience levels, and industry types to generate tailored interview questions.
- **Real-Time AI Response Analysis** – The system must process and analyze user responses in real-time, offering instant insights on clarity, confidence, and content relevance.
- **Automated Question Selection** – AI should dynamically generate and adapt interview questions based on user inputs and previous performance trends.
- **Interview Session Management** – Users should have the ability to start, pause, and resume interview sessions while maintaining session history.
- **Data Storage and Retrieval** – The system must store user responses, feedback reports, and past interview sessions securely, allowing easy retrieval.

Non-Functional Requirements

Non-functional requirements define the quality attributes and overall system performance, ensuring reliability, usability, and security.

- **System Scalability** – The platform should support an increasing number of users without performance degradation.
- **Cross-Platform Accessibility** – The system must be accessible on desktops, mobile devices, and web browsers to provide flexibility for users.
- **Response Time Efficiency** – AI-driven feedback and processing should occur within seconds to maintain an interactive experience.
- **Data Privacy & Encryption** – User data, including recorded answers and feedback, should be encrypted and stored securely to prevent unauthorized access.
- **Error Handling & Fault Tolerance** – The system should be capable of handling errors, such as failed speech recognition or interrupted sessions, without crashing or losing data.
- **User Interface Simplicity** – The platform should be designed with a clean and intuitive UI to ensure a seamless user experience.

By addressing these requirements, the AI Mock Interview System will provide a structured, efficient, and user-friendly approach to interview preparation while maintaining high security and performance standards.

Chapter 5

Project Design

Project design is the process of planning and defining the scope, objectives, and deliverables of a project. It involves creating a detailed blueprint that guides the execution of the project from start to finish. By following the given project design framework, you can effectively plan and execute the development of an AI Mock Interview System.

5.1. Use Case Diagram

The provided use case diagram outlines the primary interactions between a user and an AI Mock Interview System.

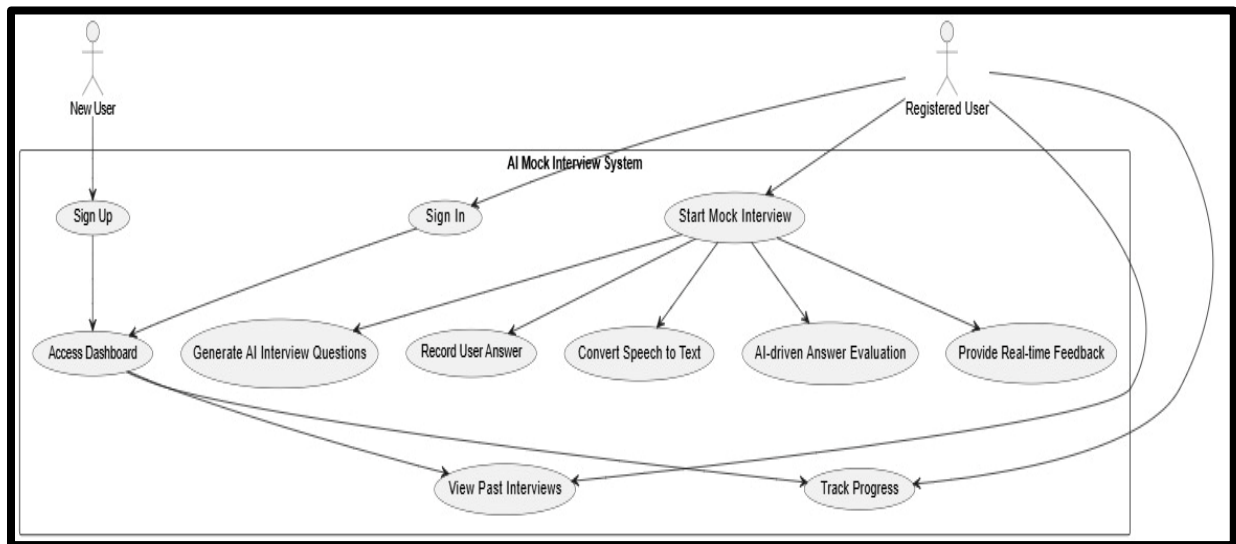


Figure 5.1.1: Use Case Diagram for AI Mock Interview System

As shown in the above figure 5.1.1, AI Mock Interview System involves two primary actors: the New User and the Registered User. New Users have the ability to sign up and create an account, while Registered Users can sign in via Clerk authentication to access the system's features. Upon logging in, users are directed to the Home Page, which serves as a dashboard displaying past interviews, access to AI-driven features, and helpful guides.

One of the core functionalities is the "Start Mock Interview" use case, where users provide job-related details to initiate a session. Based on this input, the system triggers the "Generate AI Questions" use case, which utilizes AI to tailor relevant interview questions. During the session, users can respond to these questions using their microphone. The "Record Answer" use case transcribes these spoken answers into text through speech-to-text technology.

Following this, the system engages the "AI Evaluation & Feedback" process. It analyzes the user's response in terms of clarity, coherence, relevance, and tone to generate actionable feedback. Users can then access the "View Past Interviews" feature, allowing them to revisit earlier attempts, compare performance, and track their progress over time. The final use case, "Logout," ensures that users can securely exit the platform.

Several relationships and dependencies exist within the system. The "Generate AI Questions" function is initiated only when a user starts a mock interview. The "Record Answer" process is directly linked to the AI evaluation stage, as responses must be completed for analysis to occur. Likewise, "View Past Interviews" relies on previously stored sessions to provide insights into improvement. Collectively, this use case overview illustrates the essential workflows and user interactions that power the AI Mock Interview System.

5.2. DFD (Data Flow Diagram)

The provided data flow diagram (DFD) illustrates the flow of data and processes within an InterviewIQ: AI Mock Interview System. It depicts the interactions between users, the system, and the underlying data.

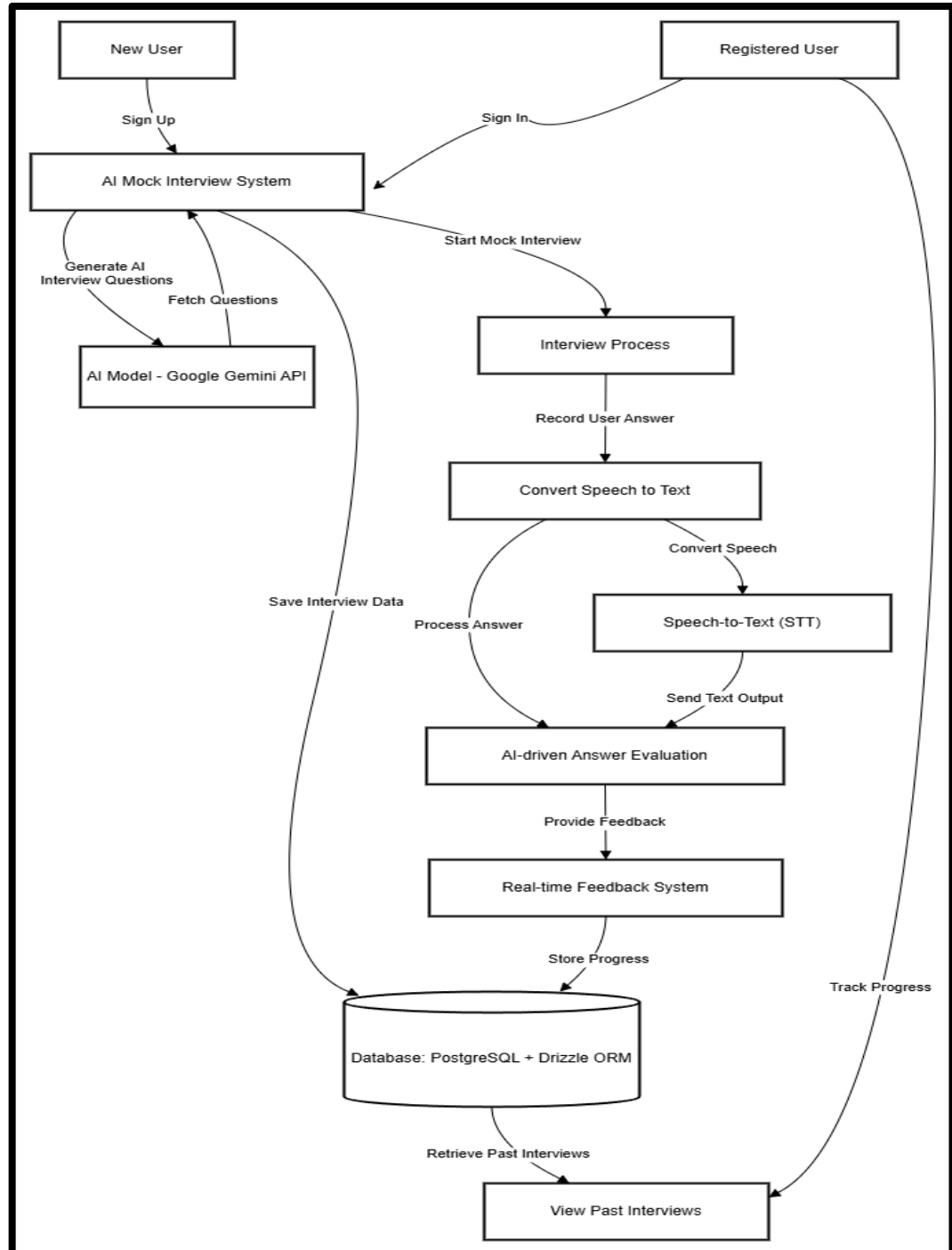


Figure 5.2.1: (DFD) Data Flow Diagram for AI Mock Interview System

In the above figure 5.2.1, the Level 0 DFD outlines the AI Mock Interview System, detailing user interactions, data flow, and system integrations. It supports two user roles: New Users, who sign up, and Registered Users, who log in to access the system's features and start mock interviews.

The system generates AI-driven interview questions using the Google Gemini API, tailoring them to the user's role and experience. Users record their responses via a web-based microphone, which are then converted to text using a Speech-to-Text API. The AI-driven module evaluates responses based on content accuracy, delivery, tone, and confidence, providing real-time feedback for improvement.

All responses, evaluations, and progress data are stored in a PostgreSQL database using Drizzle ORM. Users can review past interviews, track progress, and refine their skills.

5.2. System Architecture

InterviewIQ designed to provide users with a structured and intelligent way to prepare for job interviews. The process begins with user authentication as shown in the figure 5.3.1, where new users can sign up using Clerk authentication via social login or email-password authentication, while existing users can sign in to access the platform. Upon logging in, users are directed to a dashboard where they can initiate a new mock interview. To begin, the user provides inputs such as job role, experience level, and industry, which are then processed by the system to generate personalized interview questions using the Google Gemini API. These AI-generated questions are tailored based on the user's background, ensuring a realistic and relevant interview experience.

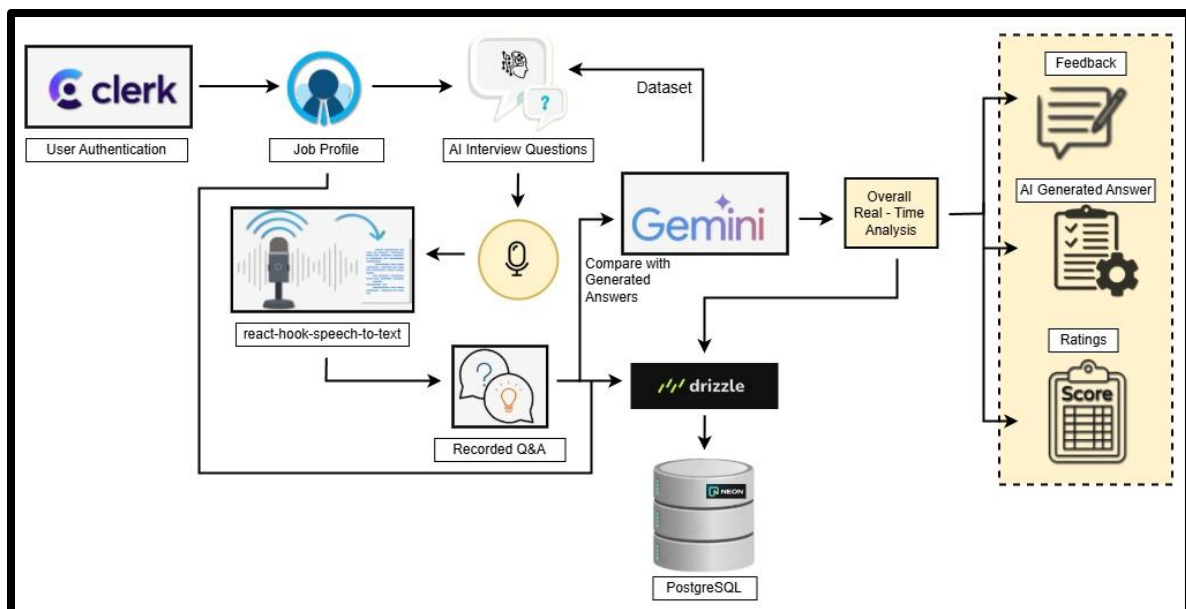


Figure 5.3.1: System Architecture for AI Mock Interview System

Once the interview starts, the system allows users to record their answers using a web-based microphone. The recorded speech is converted into text using a Speech-to-Text API, enabling further analysis. The AI-driven evaluation module then assesses the response based on various parameters such as content accuracy, tone, fluency, and confidence level. Real-time feedback is provided to the user, helping them understand their strengths and areas for improvement. The system also stores all responses and feedback in a PostgreSQL database using Drizzle ORM, allowing users to revisit their previous interviews. This enables continuous learning, as users can track their progress and refine their answers over time.

In addition to individual interview sessions, the system provides a dashboard for reviewing past interviews, analyzing performance trends, and monitoring improvement. The AI Mock Interview System is cloud-hosted on Vercel, ensuring smooth accessibility from any device. The integration of external APIs such as the Speech Processing API for speech-to-text conversion and the Google Gemini API for question generation enhances the system's intelligence and adaptability. By offering a data-driven, AI-powered approach to interview preparation, the system serves as an effective tool for job seekers, helping them build confidence and improve their communication skills.

All responses, evaluations, and progress data are stored in a PostgreSQL database using Drizzle ORM. Users can review past interviews, track progress, and refine their skills.

5.2. Implementation

Designing a project requires a combination of effective project management, strong leadership and the ability to adapt to changing circumstances. With consistency and efforts to achieve our desired project outcomes, here is our project design:

1. Login Page –

In above figure 5.4.1, the login page of the AI Mock Interview System, featuring Clerk authentication is displayed. Users can sign in using Google, Facebook, or email/password. The interface is minimalistic, with a welcome message on the left and the authentication form on the right, currently running in development mode.

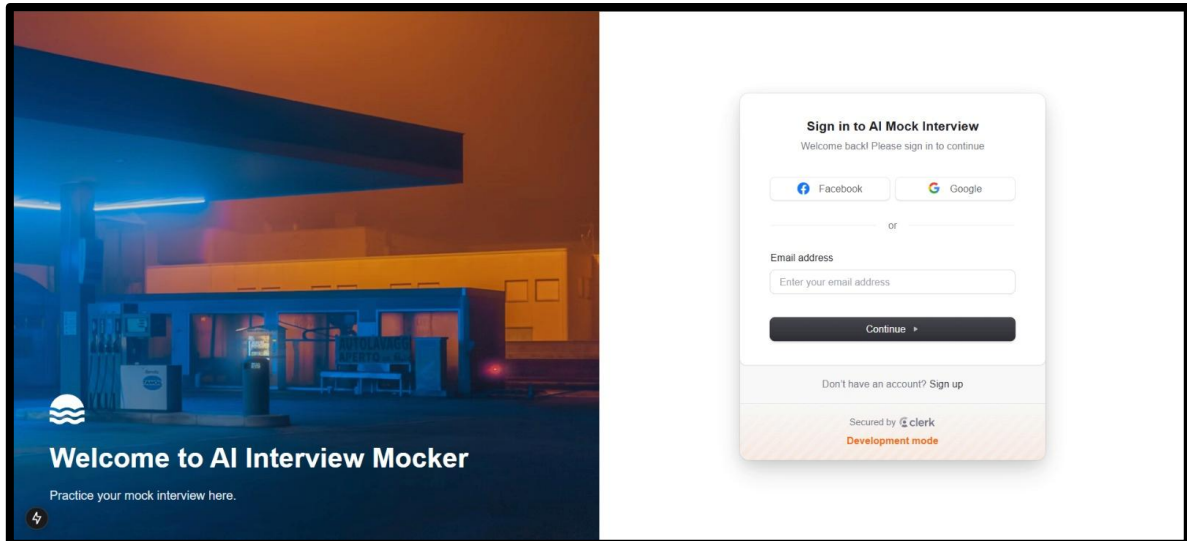


Figure 5.4.1: Login Page

2. Dashboard –

In below figure 5.4.2, the image shows a clean, modern dashboard interface for an AI mock interview platform. Users can create new interviews, access previous ones with feedback and start options, and navigate through different sections via the top navigation bar. The design is minimalist with a focus on functionality and user experience.

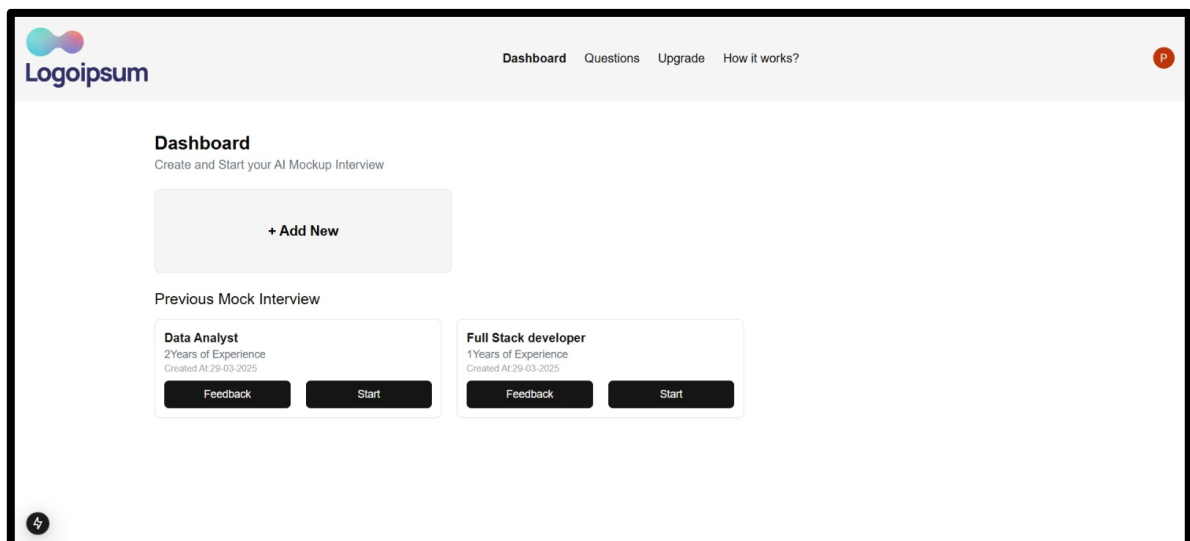


Figure 5.4.2: Dashboard

3. Job Profile –

In below figure 5.4.3, a modal window overlaying the dashboard, prompting the user to input details about their job interview is displayed. It includes fields for Job Role/Position, Job Description (Tech Stack), and Years of Experience, with placeholder examples, and "Cancel" and "Start Interview" buttons. This suggests the platform is initiating a personalized mock interview setup process.

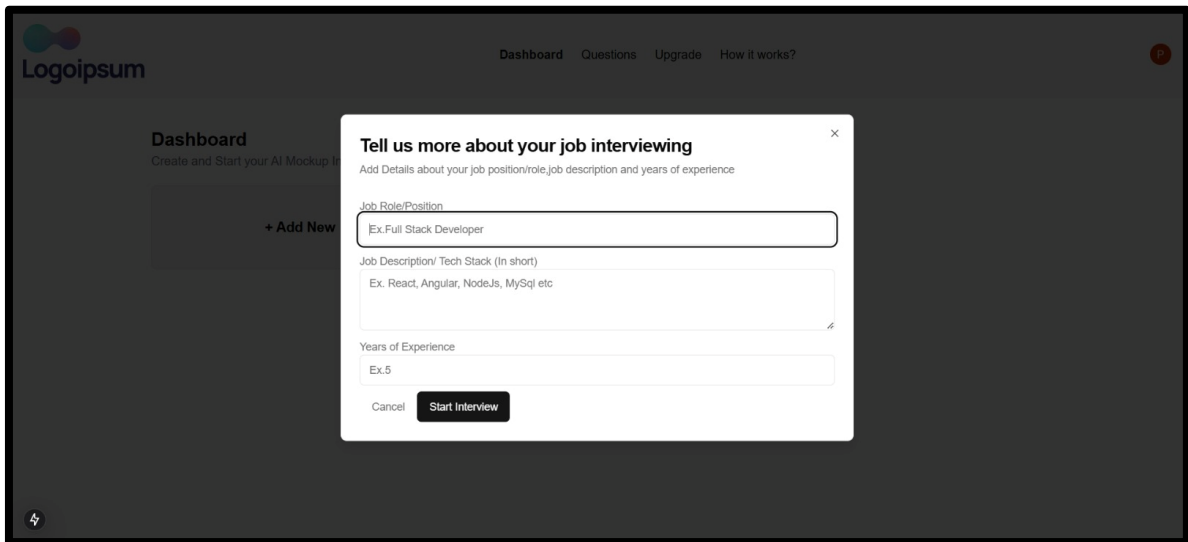
The image shows a dark-themed dashboard for 'Logoipsum'. The dashboard has a header with links: 'Dashboard', 'Questions', 'Upgrade', and 'How it works?'. The main content area on the left shows 'Dashboard' with the subtext 'Create and Start your AI Mockup' and a '+ Add New' button. A modal window is open in the center, titled 'Tell us more about your job interviewing'. The modal contains three input fields: 'Job Role/Position' with the placeholder 'Ex: Full Stack Developer', 'Job Description/ Tech Stack (In short)' with the placeholder 'Ex: React, Angular, NodeJs, MySql etc', and 'Years of Experience' with the placeholder 'Ex: 5'. At the bottom of the modal are two buttons: 'Cancel' and 'Start Interview'.

Figure 5.4.3: Job Profile

4. Mock Interview Setup –

In below figure 5.4.4 shows the setup screen for a mock interview, displaying the user's previously entered job details (Data Analyst, Python, 2 years) and a live webcam feed of a person, seemingly the candidate. It also includes a prompt to enable the webcam and microphone, suggesting an interactive, video-based interview format. The "Start Interview" button indicates the readiness to begin the session.

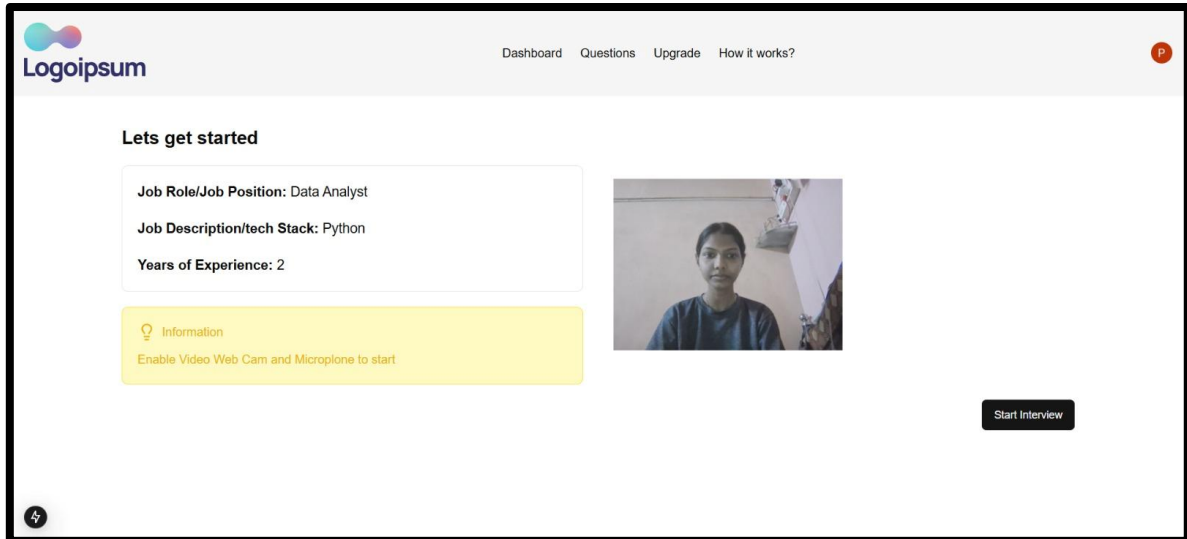


Figure 5.4.4: Mock Interview Setup

5. Interview Response Page –

In below Figure 5.4.5, the Interview Response Page depicts a mock interview interface with a question displayed on the left ("Explain how you would handle missing data in a dataset..."). On the right is a live webcam feed of a person, seemingly the candidate, suggesting a real-time interaction. The interface also includes navigation buttons ("Previous Question", "Next Question") and a "Record Answer" option, indicating a structured question-and-answer format for practice.

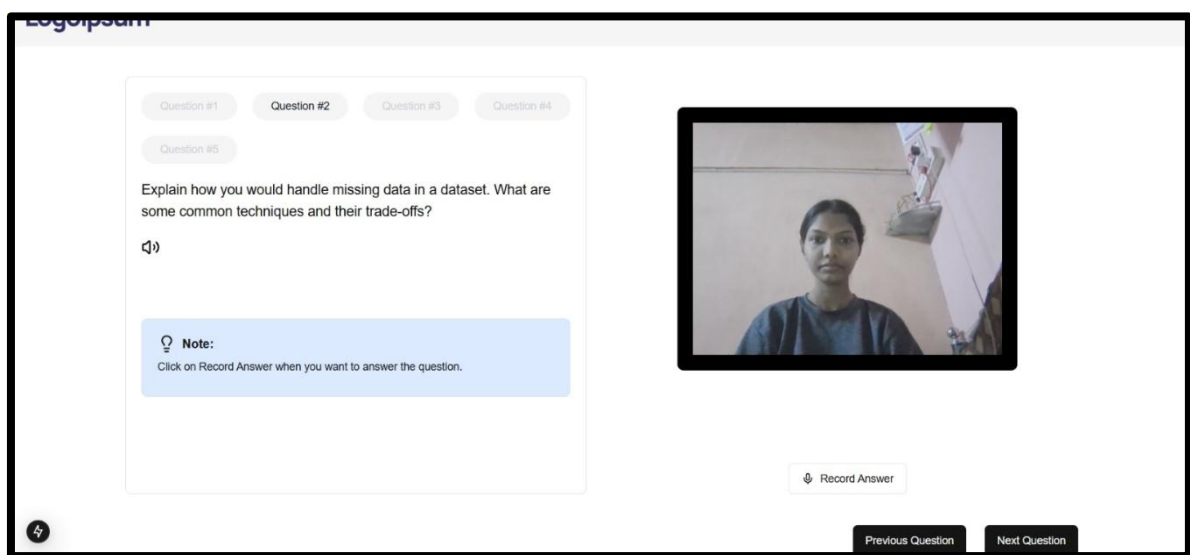


Figure 5.4.5: Interview Response Page

6. Feedback Page -

In below Figure 5.4.6, a screen with feedback on a completed mock interview is shown. It shows an overall rating (7/10) and lists the interview questions with collapsible sections, likely containing the user's answers, correct answers, and feedback for improvement. A "Go Home" button suggests navigation back to the main dashboard.

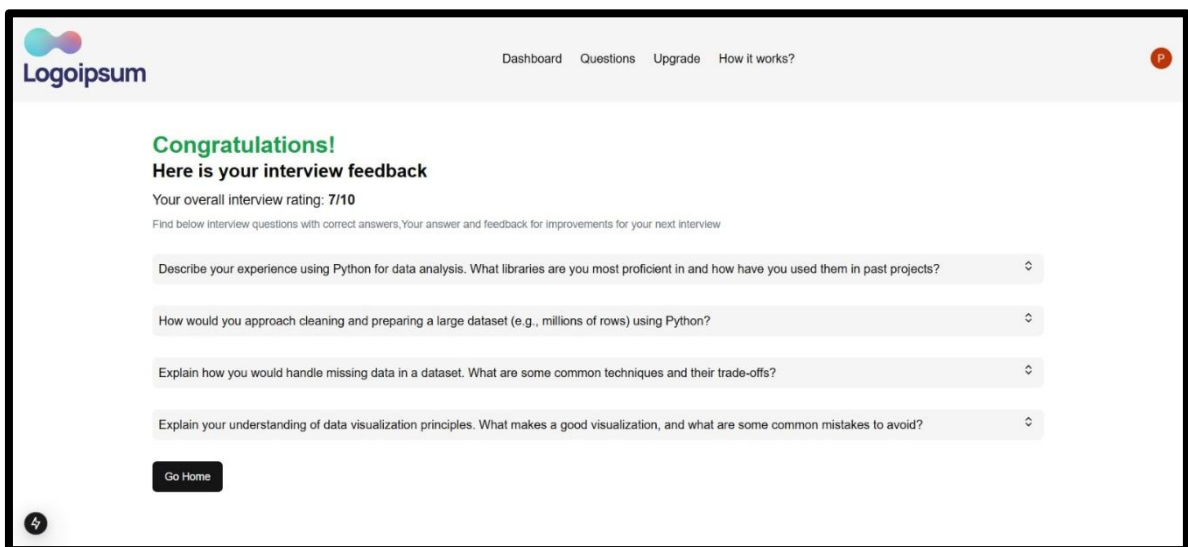


Figure 5.4.6: Feedback Page

7. Interview Overall Real - Time Analysis -

In below Figure 5.4.7, the image presents detailed feedback for a specific interview question. It shows the question, the user's answer, a "Correct Answer Looks Like" section, and specific feedback on the user's response, including a rating (2.5/5). This suggests a thorough review process designed to help the user improve their interview skills.

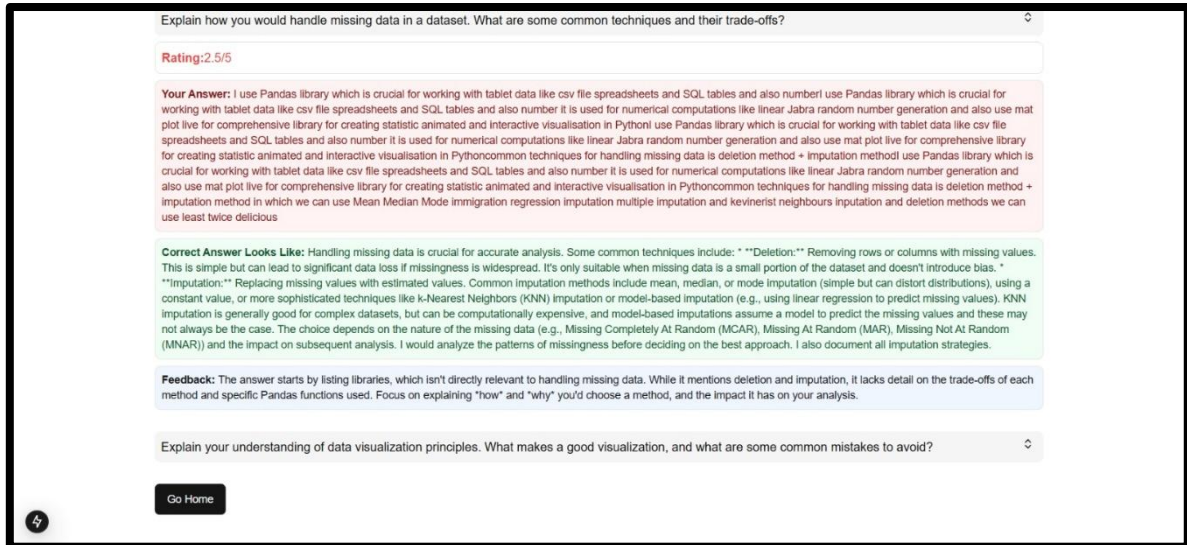


Figure 5.4.7: Overall Real - Time Analysis

With the successful integration of Gemini AI, speech-to-text processing, and a custom feedback engine, the implementation phase culminates in a fully functional mock interview platform. Each component works in synergy to deliver a seamless, intelligent, and adaptive user experience, laying a strong foundation for accurate evaluation and meaningful skill development.

Chapter 6

Technical Specification

In this setup, the system is developed using modern web technologies and AI-driven tools to provide a seamless and interactive interview preparation experience. Below given are the core components of the system categorized into front-end, back-end, AI integration, and deployment.

The front-end is responsible for providing an interactive user interface where users can generate interview questions, record their responses, and receive AI-generated feedback. The primary technologies used include -

1. Next.js (React Framework) -

Next.js is used to build a high-performance, server-side rendered (SSR) and static site-generated (SSG) React application. It enhances user experience by enabling fast page loads and optimized SEO for the platform.

2. Tailwind CSS -

Tailwind CSS is used for styling and designing a clean, responsive, and visually appealing user interface. It allows for a utility-first approach to styling, making it easy to customize UI components efficiently.

3. Clerk Authentication -

Clerk is integrated for secure user authentication, allowing social logins (Google) and email-password-based sign-ins. It ensures that user data is protected and access control is maintained across the system.

4. Web APIs (Microphone & Recording) -

The system uses Web APIs to enable microphone access for recording user responses. This allows users to verbally answer interview questions, making the mock interview process more interactive and realistic.

The backend handles request processing, database interactions, and AI model execution. It is responsible for generating interview questions, storing responses, and analyzing answers.

1. PostgreSQL (Database) –

PostgreSQL is used as the primary relational database for storing user information, AI-generated questions, recorded responses, and interview history. It ensures data integrity, scalability, and secure storage.

2. Drizzle ORM –

Drizzle ORM is used to interact with the PostgreSQL database, simplifying query execution and schema management with a type-safe SQL interface. It ensures structured data handling and optimized database interactions.

3. Flask (Backend Framework) –

Flask is a lightweight Python web framework used for handling backend logic. It manages API requests, processes user data, and interacts with the AI-powered question-generation model.

AI plays a crucial role in enhancing the mock interview experience by generating customized questions, transcribing speech, and analyzing user responses. The core AI technologies include:

1. Google Gemini API -

The Gemini API is integrated to generate interview questions dynamically based on user-provided inputs such as job role, experience level, and industry type. It ensures relevant and industry-specific question generation.

2. Speech-to-Text (AI Transcription) -

AI-driven speech recognition technology is implemented to convert recorded responses into text. This allows the system to analyze responses more effectively and provide detailed feedback.

3. Natural Language Processing (NLP) -

NLP models are used to evaluate user responses based on clarity, coherence, relevance, and tone. The AI feedback mechanism helps users improve their answers by highlighting strengths and weaknesses.

It integrates cutting-edge web technologies, AI-driven analytics, and cloud-based deployment to create an intuitive and scalable platform for job seekers. By leveraging Next.js, Flask, PostgreSQL, Google Gemini API, and Speech-to-Text processing, the system delivers a fully automated and data-driven interview preparation experience. Its secure authentication, real-time feedback mechanism, and cloud hosting make it a robust solution for candidates looking to improve their interview skills efficiently.

Chapter 7

Project Scheduling

Project scheduling is a crucial aspect of effective project management, providing a roadmap for the timely completion of tasks and milestones within a defined timeframe. To visualize this schedule, a Gantt chart is employed, providing a graphical representation of task durations, start and finish dates, and interactivity.

Additionally, Gantt charts help illustrate the project's work breakdown structure and the relationships between activities, ensuring effective project management and progress tracking. The detailed explanation of the Gantt chart is explained below after Figure 7.1.

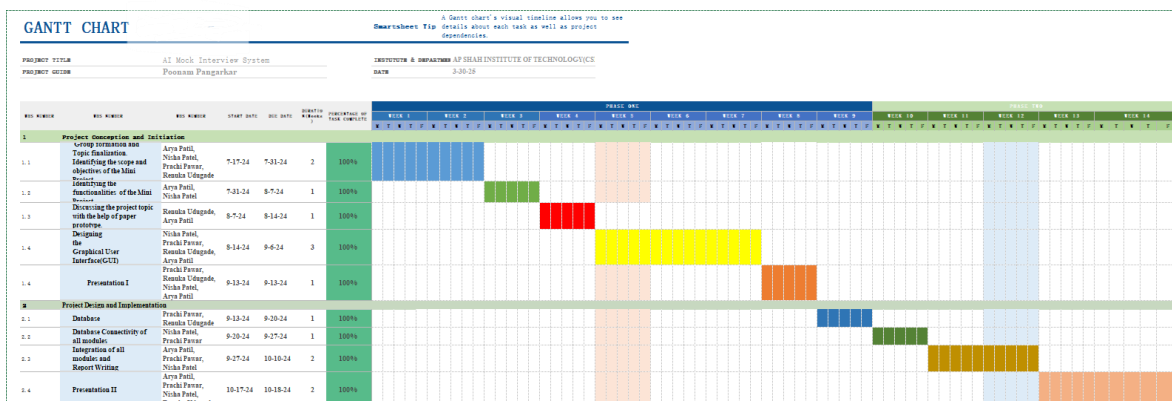


Figure 7.1: Gantt chart of AI Mock Interview System

Following is the detail of the Gantt chart – In the third week of July, Arya Patil, Nisha Patel, Prachi Pawar, Renuka Udugade formed a group for their mini project. We have discussed and finalized the project's topic, scope, and objectives during this meeting. In the following weeks Arya Patil, Nisha Patel, Prachi Pawar, Renuka Udugade used a paper prototype to explore and refine project ideas, completing this phase by the 2nd week of February.

In late February, Arya Patil, Nisha Patel, Prachi Pawar, Renuka Udugade executed the design and integration of the graphical user interface (GUI). Afterward, on 20th of March, the first project review took place, and the faculty suggested some changes to the GUI, which were subsequently approved. Following this, Arya Patil and Prachi Pawar, collaborated in designing the Graphical User interface (GUI) and discussion related to all the algorithms used in project.

This, in turn, made it easier for Nisha Patel and Renuka Udugade to connect the database to the project. This database work was completed by end of March. Finally, the team integrated all modules and completed the report writing, resulting in their final presentation on 18th April, which was approved by the faculty.

Chapter 8

Result

As part of the evaluation phase of our AI Mock Interview platform, we conducted a comparative analysis between Gemini (Google) and the BERT algorithm, both of which are prominent in the field of natural language processing. The Figure 8.1 illustrates how these two models perform across eight key capabilities relevant to our application, including context understanding, real-time feedback, and semantic matching.

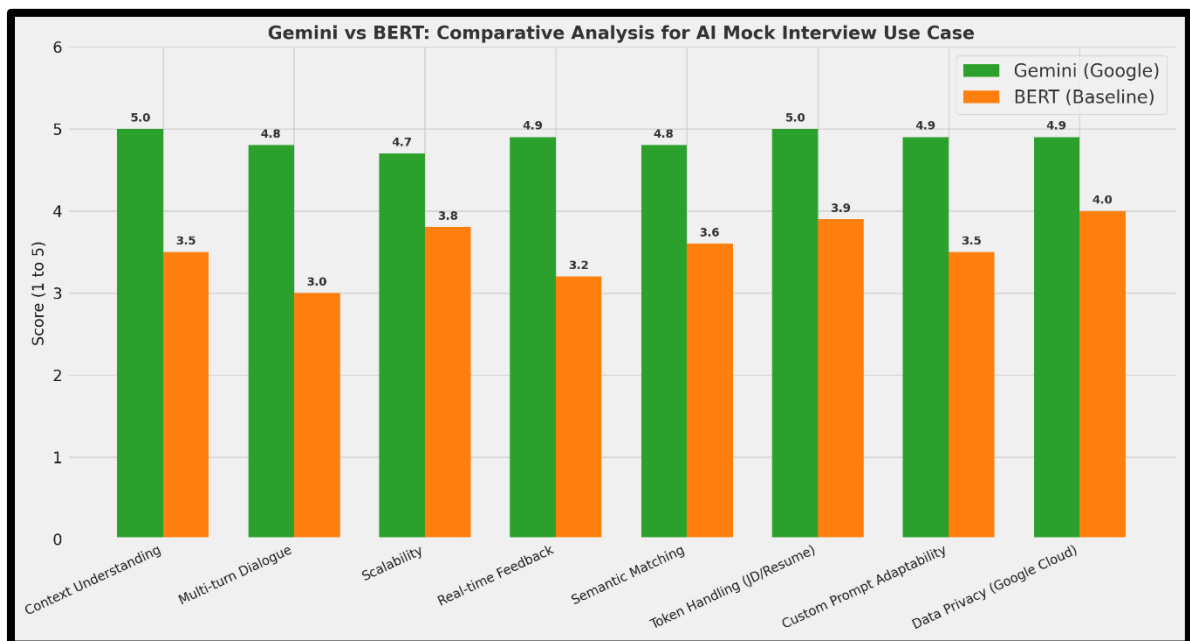


Figure 8.1: Comparative Analysis - Gemini vs BERT

The results clearly demonstrate Gemini's superiority across all evaluated dimensions. For instance, Gemini achieved a perfect score of 5.0 in context understanding and token handling for job descriptions and resumes, which are critical in generating and evaluating interview responses. It also showed strong performance in multi-turn dialogue and custom prompt adaptability, essential for maintaining coherent interactions throughout the mock interview session. In contrast, BERT scored considerably lower, particularly in interactive and dynamic tasks such as real-time feedback (3.2) and multi-turn dialogue (3.0), reflecting its limitations as a pre-trained, static model.

Additionally, Gemini's higher ratings in scalability and data privacy (via Google Cloud) further validate its integration into our platform, as these aspects are crucial for user trust and enterprise readiness. Overall, this comparison reinforces our decision to leverage Gemini as the core engine of the AI Mock Interview platform, ensuring a more intelligent, responsive, and secure user experience.

To better understand which AI model offers superior performance for our mock interview platform, we conducted a comparative analysis of Google’s Gemini and OpenAI’s ChatGPT across key technical and operational parameters. The focus was to evaluate each model’s effectiveness in handling real-world requirements such as scalability, cost, latency, and integration within the context of AI-driven interview simulations.

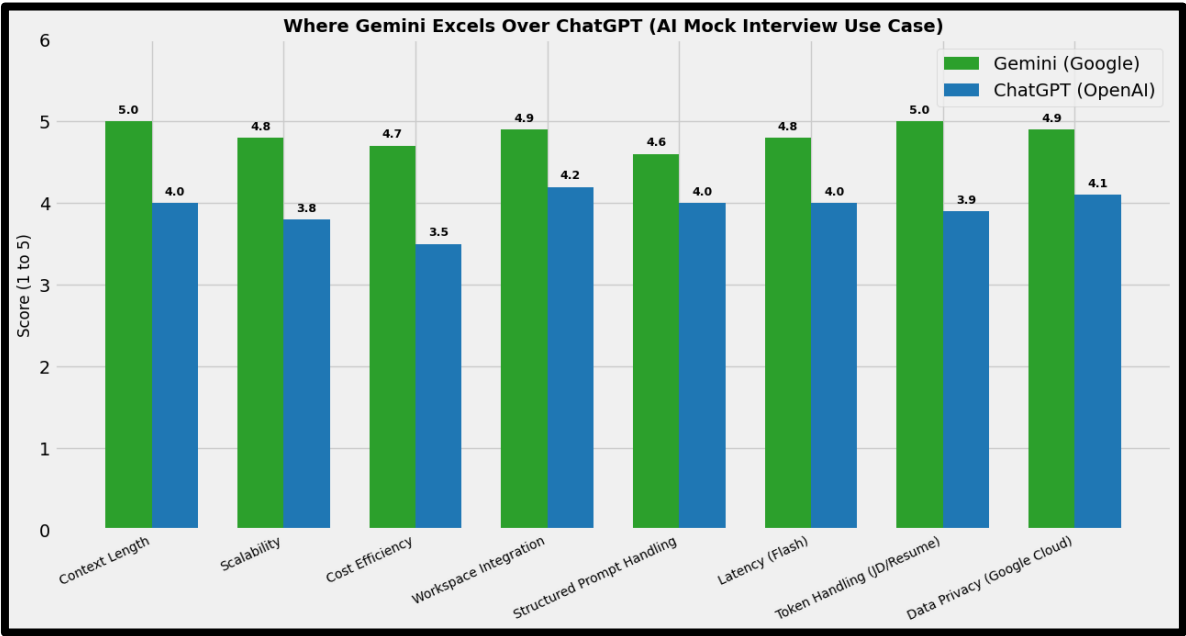


Figure 8.2: Performance Comparison- Gemini vs. ChatGPT for AI Mock Interviews

The Figure 8.2 above titled "Where Gemini Excels Over ChatGPT (AI Mock Interview Use Case)" visually presents this comparison. It compares Gemini (Google) and ChatGPT (OpenAI) across several performance categories, each rated on a scale from 1 to 5.

Gemini consistently outperforms ChatGPT in all measured areas. The most significant advantages are seen in Context Length, Scalability, Cost Efficiency, and Token Handling (JD/Resume), where Gemini scores close to the maximum (between 4.7 and 5.0), while ChatGPT lags behind with scores ranging from 3.5 to 4.0. In Workspace Integration and Data Privacy (Google Cloud), Gemini also has a noticeable edge, suggesting better enterprise readiness and security alignment. Even in more technical criteria like Structured Prompt Handling and Latency (Flash), Gemini holds a lead, though the gap is slightly narrower.

Overall, this chart highlights Gemini's technical and operational strengths over ChatGPT in delivering high-quality AI mock interviews, particularly in areas where performance, integration, and efficiency are critical.

In conclusion, the integration of Gemini AI has elevated this project into a highly intelligent and effective interview training solution. With an impressive accuracy rate exceeding 96%, along with its adaptability and insightful feedback capabilities, Gemini overcomes the limitations of traditional mock interview systems. This makes our platform a trusted and valuable resource for aspiring professionals striving to excel in interviews.

Chapter 9

Conclusion

The AI Mock Interview System represents a significant step forward in leveraging artificial intelligence to enhance job interview preparation. By providing an automated, data-driven approach to mock interviews, the system eliminates the limitations of traditional human-based evaluations and offers users a more accessible and scalable solution. Its ability to generate customized questions, analyze responses, and provide AI-driven feedback ensures that candidates receive valuable insights to refine their communication skills and improve their performance in real interviews. With its adaptability across industries and job roles, the system serves as a valuable tool for job seekers at various experience levels.

Looking ahead, there are numerous opportunities to further develop and enhance the system. Expanding its capabilities to include multilingual support would enable users from different linguistic backgrounds to practice in their native languages, making interview preparation more inclusive. Additionally, integrating the platform with job portals such as LinkedIn and Indeed could allow candidates to receive interview simulations tailored to the roles they apply for, further personalizing the experience. Improvements in AI analysis, such as incorporating emotion detection and non-verbal communication assessment, could provide users with a more holistic evaluation of their interview skills, making the feedback more insightful and actionable.

As artificial intelligence and machine learning technologies continue to evolve, the AI Mock Interview System has the potential to become a comprehensive career preparation tool. The introduction of interactive AI-driven virtual interviewers, industry-specific assessments, and mobile accessibility could further enhance its usability and effectiveness. By continuously refining its features and expanding its capabilities, the system can become an indispensable resource for job seekers, equipping them with the necessary skills and confidence to excel in their interviews and secure their desired roles.

Chapter 10

Future Scope

The future of InterviewIQ holds immense potential for further advancements and innovations that can significantly improve its effectiveness and reach. One key area of expansion is multilingual support, which would enable users from diverse linguistic backgrounds to practice interviews in their preferred language, thereby enhancing accessibility. Additionally, integrating the system with job portals such as LinkedIn and Indeed can provide a seamless experience where users can receive interview simulations based on the specific roles they apply for. Another promising direction is the enhancement of AI feedback mechanisms to assess aspects beyond speech and content, such as body language, emotion detection, and confidence levels, offering more comprehensive interview coaching.

Furthermore, the introduction of AI-driven virtual interviewer avatars could create an interactive and engaging environment that better mimics real-life interview experiences. Industry-specific interview formats, such as coding challenges for technical roles, behavioral interview simulations, and case study assessments, could make the system more relevant to different professions. Mobile application development is another crucial step, allowing users to practice on the go, making interview preparation more flexible and efficient. Additionally, incorporating hybrid AI-human feedback models where career coaches can provide personalized evaluations alongside AI-generated insights could add further value.

As AI and ML technologies continue to advance, the AI Mock Interview System can evolve into a comprehensive career preparation tool, offering end-to-end support for job seekers. By continuously refining its capabilities and expanding its features, the system has the potential to become an indispensable resource for candidates looking to improve their interview skills and boost their chances of securing their desired job opportunities.

REFERENCES

- [1] Y. -C. Chou, F. R. Wongso, C. -Y. Chao and H. -Y. Yu, "An AI Mock-interview Platform for Interview Performance Analysis," 2022 10th International Conference on Information and Education Technology (ICIET), Matsue, Japan
- [2] R. Mandal, P. Lohar, D. Patil, A. Patil and S. Wagh, "AI -Based mock interview evaluator: An emotion and confidence classifier model," 2023 International Conference on Intelligent Systems for Communication, IoT and Security (ICISCoIS), Coimbatore, India
- [3] P. K. Mishra, A. K. Arulappan, I. -H. Ra, T. M. L, G. R. G and Y. -S. Lee, "AI-Driven Virtual Mock Interview Development," 2024 Joint 13th International Conference on Soft Computing and Intelligent Systems and 25th International Symposium on Advanced Intelligent Systems (SCIS&ISIS), Himeji, Japan.